

SUBJECT: Digital Signal Processing

Exam Date: 2026-12-15

UNIT I: Discrete-Time Signals and Systems

Weightage: 20%

Topics:

- Introduction to discrete-time signals
- Classification of signals
- Time-domain representation
- Basic system properties
- Convolution sum

UNIT II: Z-Transform

Weightage: 20%

Topics:

- Definition of Z-transform
- Properties of Z-transform
- Inverse Z-transform
- Region of convergence
- Applications of Z-transform

UNIT III: Discrete Fourier Transform

Weightage: 25%

Topics:

- DFT definition
- Properties of DFT
- FFT algorithms
- Circular convolution
- Spectral analysis

UNIT IV: Digital Filter Design

Weightage: 15%

Topics:

- FIR filter design
- IIR filter design
- Window techniques
- Frequency response analysis

UNIT V: Applications of DSP

Weightage: 20%

Topics:

- Speech processing
- Image processing
- Biomedical signal processing
- Communication systems

SUBJECT: Wireless Communication

Exam Date: 2027-01-20

UNIT I: Cellular Concepts

Weightage: 15%

Topics:

- Frequency reuse
- Cell splitting
- Handoff strategies
- Capacity improvement

UNIT II: Mobile Radio Propagation

Weightage: 20%

Topics:

- Path loss models
- Shadowing
- Multipath fading
- Delay spread

UNIT III: Multiple Access Techniques

Weightage: 25%

Topics:

- FDMA
- TDMA
- CDMA
- OFDMA

UNIT IV: LTE and 5G

Weightage: N/A

Topics:

- LTE architecture
- 5G NR
- Network slicing
- Massive MIMO

UNIT V: Wireless Security

Weightage: 40%

Topics:

- Authentication
- Encryption
- Key management
- Security threats